The Data Modeling Process

30-Minute Hands-On Exercise | Week 9 - Section 2A

Learning Objectives: Define a data model and explain why modeling choices matter | List and sequence the data modeling stages in Power BI | Distinguish between OLTP and OLAP systems | Identify what makes a source an ideal starting point for a Power BI model

Instructions: Complete all four parts in 30 minutes. Refer to your Section 2A notes. Be ready to discuss your answers with the class.

Part 1: Fill in the Blank (~7 minutes)

Complete each sentence with the correct term or phrase from Section 2A.

1.	A data model describes the ___shape and relationships___ of your data and how each part relates to the others.
2.	The first stage of the modeling process is to ___discover___ the source data by learning its existing shape and which values will be useful.
3.	In relational databases, the relationship between two tables is frequently recorded using a ___foreign key___ .
4.	A database optimized for reporting, structured with star or snowflake schema, and commonly called a Data Warehouse is known as an ___OLAP___ system.
5.	The ideal Power BI model organizes data into a ___star or snowflake___ schema.
6.	Adding calculations, hierarchies, and date attributes to the model is called the ___augment___ stage.

Part 2: Matching (~6 minutes)

Write the letter of the correct description next to each term.

Term	
A. Data Model	5.
B. OLTP System	6.
C. OLAP System	3.
D. Source Model	1.

Description
1. The original shape of data before it has been modeled or transformed
2. A constraint in relational databases that points to a related row in another table
3. A database structured for reporting; less normalized, uses star/snowflake schema
4. The ideal target shape for a Power BI semantic model

E. Destination Model 4.		5. The shape that describes relationships within a dataset and how each part relates
F. Foreign Key 2.		6. Processes many simple, fast transactions; highly normalized; multi-user; emphasizes data integrity

Part 3: Modeling Stages in Order (~8 minutes)

The six data modeling stages are listed below in order. For each stage, write at least two key actions or questions that belong to that stage in the right-hand column.

Order	Modeling Stage	Key Action(s) in this Stage
1	1. **Discover**	learn shape + identify
2	2. **Bring In (Ingest)**	Verify security rights + determine time range
3	3. **Profile**	Look at source data + check for missing values
4	4. **Transform**	Combine/split tables + rename columns and tables
5	5. **Clean**	Remove/repair data + fix typos, implausible values and abnormalities
6	6. **Augment**	Add calculations low-level totals/formatted variations + define hierarchies, date attributes, pull parent/child data

Part 4: Short Answer (~9 minutes)

Answer each question in 2-4 sentences in your own words.

Question 1

A junior analyst receives a sales CSV file and immediately starts building visuals. What critical step(s) of the modeling process are they skipping, and why does that matter?

The analyst is skipping Discover, Profile, Clean, and Augment — essentially jumping straight from raw data to visuals without understanding or preparing it first. This matters because if the data contains bad values, wrong data types, or missing relationships, the visuals built on top of it will produce incorrect or misleading results. You can't trust a report built on data you haven't profiled and cleaned — garbage in, garbage out.

Question 2

Your source data uses the data type 'Float' for a price column. You want report users to see values displayed as currency (e.g., $1,200.00). At which modeling stage would you address this, and why is Power BI better suited for it than a traditional relational database?

This would be addressed during the Transform stage, specifically when defining column data types and formatting. Power BI is better suited for this than a traditional relational database because relational databases only support generic numeric types like Float or Decimal, while Power BI's modeling layer supports human-oriented types like Currency, Percentage, and custom date formats that are designed specifically to make reports readable and meaningful to business users.

Question 3

What is the difference between a 'simple' calculation and an 'advanced' calculation in the Augment stage? Give a concrete example of each.

A simple calculation involves data from a single row or a straightforward lookup. For example, multiplying [Quantity] by [Unit Price] to get a row-level [Line Total]. An advanced calculation considers data from many rows or across different filter contexts. For example, calculating what percentage of total annual revenue a single product line represents, which requires summing across rows and evaluating the result relative to a broader group.

Question 4

Your company's data lives in a highly normalized OLTP database with 40 tables. A colleague suggests connecting Power BI directly to it without any modeling changes. What two problems could arise, and what would be a better approach?

Two problems that could arise: first, the 40 highly normalized tables likely have circular relationships that Power BI can't handle cleanly, which would break the model or produce incorrect filter behavior. Second, OLTP tables are designed for fast transactions, not analysis — the structure makes it very difficult to write clean DAX measures and build intuitive reports. A better approach would be to use Power Query to reshape and consolidate those 40 tables into a proper star schema with clearly defined data tables and lookup tables before building any visuals on top.

Bonus / Reflection (if time allows)

Section 2A states: 'Understanding your source data is essential to the modeling process -- and understanding an ideal destination model is just as essential.' In 3-5 sentences, explain what this means in practice. Why is it not enough to understand just one side?
